

Adapting Large Language Models for Biomedicine through Retrieval-Augmented Generation with Documents Scoring

Yinkui Huang[†], Tianrun Gao[†], Jiangjiang Zhang[†], Xiaohong Liu[§], Guangyu Wang^{†*}

[†] State Key Laboratory of Networking and Switching Technology,
Beijing University of Posts and Telecommunications, Beijing, China

[§] UCL Cancer Institute, University College London, London, United Kingdom

realfawkes@bupt.edu.cn, tianrun.gao@bupt.edu.cn,
jiangjiang02662003@gmail.com, xhliu17@gmail.com,

*Corresponding author: Guangyu Wang

Corresponding author email: guangyu.wang@bupt.edu.cn

Abstract—By integrating the generative capabilities of Large Language Models (LLMs) with external biomedical knowledge repositories, Retrieval-Augmented Generation (RAG) offers significant potential for addressing knowledge-intensive biomedical tasks, enhancing the precision and effectiveness of clinical decision-making. However, the standard RAG approach fails to fully leverage the information within the retrieved documents, which can lead to incorrect response generation. While advanced RAG approaches empower LLMs to understand the correlation between queries and documents, they require costly annotation. In this work, we propose a strategy to help LLMs better leverage the information within retrieved documents in RAG system while reducing data acquisition costs. We fine-tuned an LLM on a dataset comprising queries, retrieved documents, and their relevance scores generated by a pre-trained biomedical re-ranker to adapt the model for answering questions from reference documents. The fine-tuned model can independently score retrieved documents before answering the question, thereby more effectively utilizing the information contained within retrieved documents. Experimental results shows that we have improved the model's accuracy on two biomedical benchmarks while reduced the costs of data acquisition.

Index Terms—Large Language Models, Retrieval-Augmented Generation, score documents

I. INTRODUCTION

Large Language Models (LLMs) have demonstrated noteworthy success in various biomedical applications [1] [2] [3]. However, LLMs face challenges when simply prompted to address domain-specific biomedical tasks. These models often exhibit errors, termed as hallucinations, when relying solely on their internal parametric knowledge to handle knowledge-intensive tasks [4]. Enhancing the capability of LLMs to handle such tasks effectively could improve the precision and effectiveness of clinical decision-making [5].

Retrieval-Augmented Generation (RAG) is an approach to address the hallucination issue in LLMs. It enhances their performance by retrieving pertinent documents from external repositories and integrating them into the generation process, thereby improving performance in knowledge-intensive tasks

[5] [6]. In the biomedical domain, integrating the generative capabilities of LLMs with external biomedical knowledge repositories, such as PubMed and PMC, makes RAG advantageous and better suited for coping with biomedical knowledge-intensive tasks [7].

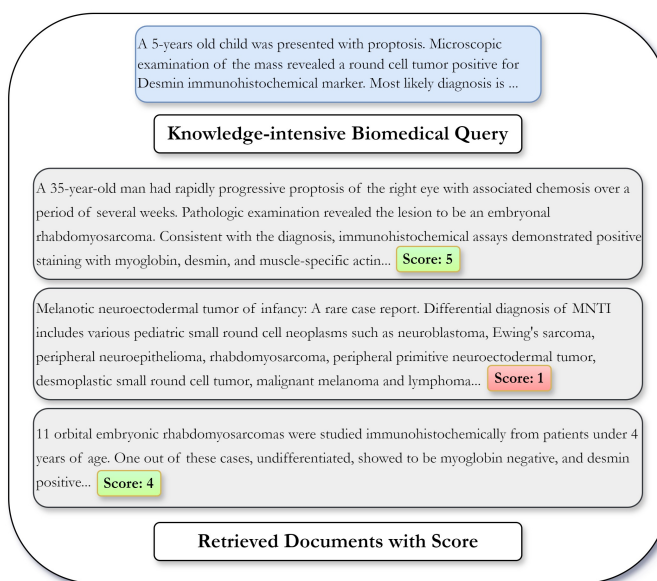


Fig. 1. Example of query and retrieved documents with score.

We argue that the retrieved biomedical documents in RAG system contain substantial knowledge helpful for answering knowledge-intensive questions. However, the standard RAG approach merely inputs documents into the LLMs to generate an answer without fine-tuning them, failing to help the LLMs understand the correlation between queries and documents or fully leverage the information contained in the retrieved documents. Although advanced RAG approaches fine-tune LLMs with labeled documents to help them understand the

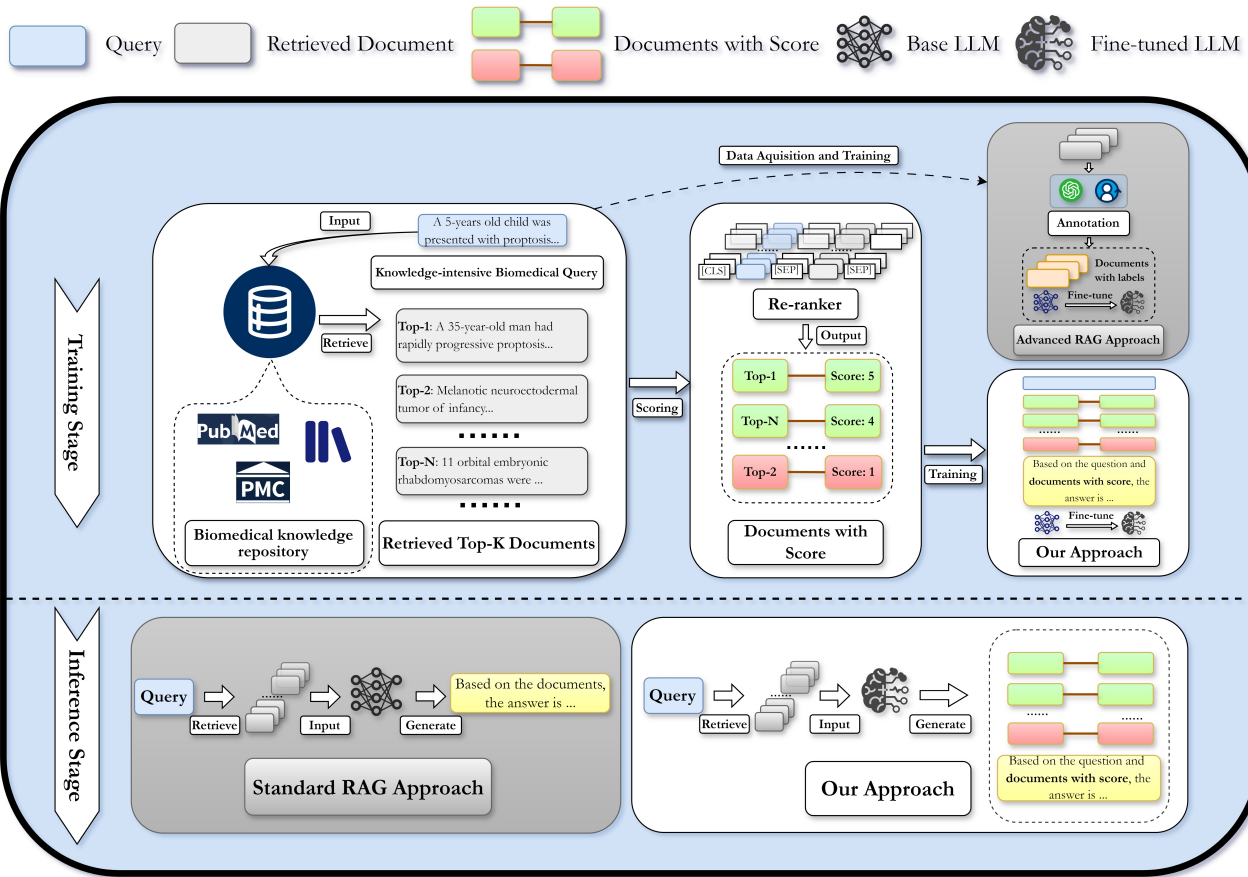


Fig. 2. Overview of adapting LLMs for biomedicine through RAG with documents scoring.

correlation between queries and documents, these methods require either supervised datasets, synthetically generated additional instructions, or GPT-4 [8] labeled instruction-tuning data, all of which lead to expensive annotation or increased complexity [9] [10] [11].

Consequently, to adapt LLMs for better leverage the information within the retrieved documents while reducing data acquisition costs, we propose a strategy to enhance Large Language Models for biomedical Retrieval-Augmented Generation by empowering them with the ability to score retrieved documents before answering knowledge-intensive questions. Specifically, we use a pre-trained, domain-specific re-ranking model (i.e., re-ranker) to generate relevance scores between queries and documents in training stage. The re-ranker prioritizes retrieved documents based on their relevance with query, considering the correlation between query and documents [12] [13]. Examples of documents with their scores are illustrated in Fig. 1. Afterward, we fine-tune the LLM on a dataset consisting of queries, retrieved documents, and their relevance scores. The fine-tuned model is capable of scoring documents independently before answering the knowledge-intensive question, thus more effectively utilizing the information within retrieved documents before answering questions.

In general, our contributions can be summarized as follows:

- We propose a domain adaptation approach for LLMs in biomedicine to address knowledge-intensive tasks by empowering LLMs to score documents within the RAG system, thereby more effectively leveraging the information within retrieved documents.
- We fine-tune an LLM on a dataset comprising queries, retrieved documents, and their relevance scores generated by a pre-trained biomedical re-ranker. The fine-tuned LLM can independently score retrieved documents during the inference stage before answering knowledge-intensive questions.
- Experimental results demonstrate that we improved the model's accuracy on two biomedical benchmarks while reducing data acquisition costs.

II. RELATED WORKS

The standard RAG approach has been recognized as a valuable tool for addressing knowledge-intensive tasks [5] [6]. It inputs query and retrieved documents to LLMs for generate an answer without fine-tuning them, failing to enable the LLMs to understand the correlation between queries and documents. Advanced RAG approaches fine-tune LLMs with labeled retrieved documents along with input query to help the models understand the correlation between them. Specifically, Self-RAG [9] proposes a RAG framework that introduces special

reflection tokens to control LLM's behavior and uses a state-of-the-art LLM GPT-4 to annotate instruction-tuning datasets. SelfBioRAG [10] inherits the framework of SelfRAG, expanding biomedical instruction-tuning datasets by leveraging Self-Instruct [14] and GPT-4 annotation. RAFT [11] combines relevant and irrelevant documents, fine-tuning the LLM without explicitly indicating which documents are irrelevant. These methods require either GPT-4 labeled customized instruction-tuning data, synthetically generated additional instructions, or supervised datasets, all of which demand costly annotation. Adapting LLMs to better leverage the information within documents to address knowledge-intensive biomedical tasks while reducing data acquisition costs has not been sufficiently investigated..

III. METHODS

A. Overview

Fig. 2. illustrates an overview of our strategy. Specifically, in the training stage, we retrieve Top-K relevant documents from the biomedical knowledge repository and get their relevance scores using a domain-specific re-ranker. Upon completing data collection, we fine-tune the LLM on a dataset comprising queries, retrieved documents, and their relevance scores. The fine-tuned LLM can independently score documents before answering knowledge-intensive biomedical questions in the inference stage.

The knowledge repository plays a vital role in the RAG system. It contains 0.1 billion snippets of biomedical documents, each with an average length of 128 words, collected from PubMed, PMC, and medical textbooks [15]. Moreover, we select MedCPT-Article-Encoder and MedCPT-Query-Encoder to encode documents and queries respectively, and MedCPT-Cross-Encoder as the re-ranker, as they have demonstrated promising performance in biomedical retrieval tasks and RAG systems [7] [12].

B. Data collection

Given the query Q , we retrieve documents in the knowledge repository. Each segment of text t_i in the repository is encoded by the embedding model to obtain its embedding representation $d_i \in \mathbb{R}^h$, where h denotes the hidden size of embedding. We save the encoded repository $\mathcal{D}$, denoted as:

$$\mathcal{D} = \{d_1, d_2, \dots, d_N\} \in \mathbb{R}^{N \times h} \quad (1)$$

where N represents the number of documents in $\mathcal{D}$.

Subsequently, Q is fed into embedding model to generate its representation $q \in \mathbb{R}^h$ and conduct a maximum inner product search between q and $\mathcal{D}$ to find the top-K most similar documents $D_1 = \{d_1^r, d_2^r, \dots, d_K^r\}$:

$$D_1 = \operatorname{argmax}_{D^* \subset \mathcal{D}, |D^*|=K} \sum_{d \in D^*} d^T q \quad (2)$$

We randomly sampled negative samples following [11]. Afterward, we apply the re-ranker to score the relevance

between q and each document candidates $d^r \in D_1$ from the previous step:

$$S_q^{d^r} = W^T T([\text{CLS}]q[\text{SEP}]d^r[\text{SEP}])_{[\text{CLS}]} + b \quad (3)$$

where T denotes the Transformer [16] encoder blocks, $[\text{CLS}]$ and $[\text{SEP}]$ are the special tokens used in BERT [17]. We use the special $[\text{CLS}]$ token as the pooled representation of the whole sentence pair to predict relevance scores. The cross-encoder relevance scores consider the correlation between query and document representation, incorporating more semantic information compared to merely measuring similarity between embeddings.

Thus far, we have obtained the training data $\mathcal{D}_{score}$

$$\mathcal{D}_{score} = \{m | m = (q, D_1, S)\} \quad (4)$$

where the relevance scores $S_q^{d_i^r} \in S$ reflect the correlation between query and documents.

We randomly sampled 200 documents to evaluate the validity and accuracy of relevance scores and achieved a 92.5% agreement with GPT-4 judgment, indicating a high level of consistency with powerful LLM's consensus on relevance scores. Furthermore, experiments in Section IV demonstrate the validity of the relevance score.

C. Training Stage

Given the input x , documents D_1 and S embraces the relevance scores, we train the generator model $\mathcal{G}$ on the dataset using standard next token objective:

$$\max_{\mathcal{G}} \mathbb{E}_{(x,y,S) \sim \mathcal{D}_{score}} \log p_{\mathcal{G}}(S, y | (x, D_1)). \quad (5)$$

where x comprises the prompt and Q , and $\mathcal{G}$ learns to predict the relevance scores along with target output sequence y . During the training stage, this strategy improves the LLM's generalizability in the biomedical domain while maintaining its text generation capabilities. Additionally, evaluating the relevance between queries and documents before generating output sequences enables the LLM to better understand their correlation, thereby leveraging the helpful information contained in the retrieved documents.

D. Inference Stage

We employ the same maximum inner product search method during data collection to retrieve biomedical documents from $\mathcal{D}$. Upon the completion of training, $\mathcal{G}$ has attained the capacity to score retrieved documents prior to answering the question using the standard next token objective:

$$f(y, D_2) = p(S, y | (x, D_2)) \quad (6)$$

Where D_2 represents the retrieved documents in the inference stage. The fine-tuned model can independently score documents, thereby more effectively leveraging the information within the retrieved documents before answering questions.

TABLE I
STATISTICS OF THE DATASETS.

Dataset	Train	Validation	Test
PubMedQA	201,000	11,269	500
MedMCQA	182,822	3,683	500

IV. EXPERIMENTS

A. Datasets

We evaluate our model and baselines on two open-domain biomedical benchmarks. PubMedQA [18] is a biomedical benchmark collected from PubMed abstracts to measure the reasoning and inferring abilities of biomedical artificial intelligence. To investigate the capability of the RAG system in identifying relevant documents and accurately answering questions based on those documents, we removed the references following [7] [19]. MedMCQA [20] is a benchmark designed to address real-world medical questions. The questions cover a wide range of 2.4k healthcare topics and 21 medical subjects. Statistics of these two datasets are summarized in Table I.

B. Baseline comparison

To better assess the competitiveness of our model, we selected LLMs with larger parameters, including Llama2_{70B} dense model [21] and Mixtral_{8x7B} MoE model [22], as well as LLMs pre-trained or fine-tuned on biomedical corpora like MedAlpaca [3] and PMC-Llama [2]. We also include GPT-3.5 [23] as a representative proprietary model. Due to the constraints of the context window, we use zero-shot prompts during the inference stage. For baseline open-source LLMs, we employ greedy search with a repetition penalty of 1.2 to ensure stable output sequences. For SelfRAG, SelfBioRAG and RAFT, we use their default settings. We opted not to use other LLMs as our base model since the baselines utilized Llama2.

C. Implementation settings

We use 4 Nvidia A100 with 80GB memory GPUs to train our model on a dataset of 383.8k samples. The model is trained for 3 epochs with a batch size of 16, a peak learning rate of 1e-5 with 10% warmup steps to prevent gradient explosion, and linear decay afterward. We set the maximum token length to be 2,048 for the model. Moreover, we deploy LoRA [24] with $LoRA_{rank} = 64$, $LoRA_{\alpha} = 128$ due to the memory constraint. A LoRA module is inserted in all layers except the embedding layer and the language modeling head layer. We use DeepSpeed stage 2 [25] to conduct multi-GPU distributed training. Since INT8 quantization [26] can not satisfy the training precision, we fine-tune our model with training precision FP16 enabled.

V. RESULTS

A. Experimental results

We evaluate our model on two biomedical benchmarks and make comparisons with various baselines. Experimental

results for benchmarks are presented in Table II and Table III, where Acc. represents the accuracy.

TABLE II
BASELINE COMPARISON ON BIOMEDICAL BENCHMARKS

Biomedical Benchmarks				
Model	Parameter	PubMedQA [18]	MedMCQA [20]	Average
		(Acc.)	(Acc.)	(Acc.)
Baselines without retrieval				
GPT-3.5 [8]	NaN	51.0	51.0	51.0
Llama2 [21]	70B	38.6	43.6	41.1
Mixtral [22]	8x7B	39.4	45.4	42.4
PMC-Llama [2]	7B	30.0	25.0	27.5
MedAlpaca [3]	7B	33.8	27.0	30.4
Baselines with retrieval				
GPT-3.5	NaN	67.4	58.0	62.7
Llama2	70B	43.6	44.4	44.0
Mixtral	8x7B	47.0	47.4	47.2
PMC-Llama	7B	29.4	23.8	26.6
MedAlpaca	7B	33.8	28.2	31.0
Ours	7B	55.6	43.6	49.6

1) *Baselines without retrieval*: Table II (top) presents the baselines without retrieval. Our 7B model (the last row) demonstrates a performance advantage over larger parameter models (70B Dense model and 8x7B MoE model), lagging only 1.4% behind GPT-3.5 in average accuracy (49.6% vs. 51.0%). Our approach also significantly outperforms 7B models pre-trained or fine-tuned on extensive biomedical corpora, indicating the efficiency of our strategy in adapting the model to domain-specific knowledge-intensive tasks.

2) *Baselines with retrieval*: Table II (bottom) presents the baseline models with retrieval using standard RAG approach. Compared to the baselines without retrieval, most models show performance improvements on the two benchmarks except PMC-Llama. It is worth highlighting that more powerful models benefit more from retrieval, where GPT-3.5 achieves 62.7%, an 11.7% increase than disable retrieval. Mixtral_{8x7B}, Llama2_{70B} and Medalpaca_{7B} gained 4.8%, 2.9% and 0.6% from retrieval respectively and PMC-Llama even decreased 0.9% when funnel retrieved documents. This suggests that retrieved documents include substantial knowledge helpful to answer the question. However, the model has not fully leveraged the information it contains, i.e., adapted to answer questions based on retrieved documents.

Compared with baselines with retrieval on different base models, our model still shows higher accuracy than all models in average accuracy except GPT-3.5. With the introduction of training LLM on retrieved documents and their relevance scores, our model has gained the ability to better harness the domain-specific knowledge in retrieved documents that contributes to answering questions.

Table III presents the baselines with retrieval using the same base model (Llama2_{7B}) with different advanced RAG approaches, in comparison to our model. We opted not to use other LLMs as our base model since the baselines utilized Llama2. In terms of average accuracy, our model achieved

TABLE III
DIFFERENT RAG STRATEGIES ON BIOMEDICAL BENCHMARKS

Model	Parameter	Biomedical Benchmarks		
		PubMedQA (Acc.)	MedMCQA (Acc.)	Average (Acc.)
<i>Llama2_{7B} with retrieval</i>				
SelfRAG [9]	7B	41.8	38.6	40.2
SelfBioRAG [10]	7B	54.6	42.0	48.3
RAFT [11]	7B	48.4	37.6	43.0
Ours	7B	55.6	43.6	49.6

49.6% on the benchmarks, which exceeds all baseline models. Among other approaches, SelfRAG achieved 40.2%(-9.4%) of average accuracy on the benchmarks. The best results of the baselines are SelfBioRAG with 54.6%(-1.0%) on PubMedQA and 42.0%(-1.6%) on MedMCQA. RAFT, achieves 48.4%(-7.2%) and 37.6%(-6.0%) on two benchmarks, which is also lower than our method. Experimental results shows that we have significantly reduced the costs of data acquisition in the training stage while maintaining the model’s competitiveness in the inference stage on biomedical knowledge-intensive tasks.

B. Ablation analysis

TABLE IV
ABLATION STUDIES FOR KEY COMPONENTS OF OUR STRATEGY

	PubMedQA (Acc.)	MedMCQA (Acc.)	Average (Acc.)
Ours	55.6	43.6	49.6
<i>Training Stage</i>			
w/o. scoring	48.0	37.2	42.6 (-7.0)
w/o. scoring + retrieval	36.2	33.6	34.9 (-7.7)
<i>Inference Stage</i>			
w/o. fine-tune	34.6	29.8	32.2 (-17.4)
w/o. fine-tune + retrieval	32.4	27.2	29.8 (-2.4)

To thoroughly examine the contribution of each component of our model, we conduct a comprehensive ablation analysis on the benchmarks. The results of these experiments are detailed in Table IV.

From the training stage perspective, both the introduction of relevance scores and the incorporation of retrieved biomedical knowledge in training data contribute to our model’s performance. Specifically, we observed that the average accuracy without document relevance scores decreases by 7.0% (42.6% vs. 49.6%), which indicates that accounting for the correlation between queries and documents and explicitly guiding the LLM during the training stage, will enhance its performance. Furthermore, after removing the relevance scores during the training stage, we further disabled the retrieval process, resulting in an additional performance decrease of 7.7% (34.9% vs. 42.6%). This demonstrates that integrating

retrieved documents into the input when fine-tuning LLM is capable of enhancing LLM’s capacity to answer questions based on documents.

Additionally, we examine the contribution of fine-tuning itself. We observed that the key to our model is the introduction of fine-tuning LLM on a dataset consisting of queries, retrieved documents, and their relevance scores. The accuracy of without fine-tuning decreases by 17.4% (32.2% vs. 49.6%) compared to the complete model. Moreover, removing the retrieval process causes a performance degradation of 2.4%, similar to the performance drop observed in most base models in previous experiments after removing external biomedical knowledge. Notably, the fine-tuned model achieved a 10.4% (42.6% v.s. 32.2%) improvement in the base model even without relevance score. This further suggests that fine-tuning the model with the concatenation of retrieved documents in input makes it more adept at answering questions based on reference documents.

VI. CONCLUSION

In this paper, we proposed a strategy to adapt Large Language Models for biomedical Retrieval-Augmented Generation by fine-tuning the model on the dataset comprising queries, documents, and relevance scores. The fine-tuned model is capable of scoring documents independently before answering the knowledge-intensive questions. Experiments on two biomedical benchmarks demonstrate that our strategy enables LLMs to better leverage the information within retrieved documents while reducing data acquisition costs. In the future, we plan to extend our approach to additional tasks and leverage different biomedical knowledge repositories.

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